

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	VR, Robotics, and "Monkey in the Middle"
<i>Company:</i>	Please provide company name
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Mixed Reality, HCI, Robotics, AI
Project Background Information	
Interacting with robots often seems to be a very sedate and serious affair, especially with dedicated control interfaces and modes of commands. Robots however are becoming increasingly commonplace in daily human life, and the traditional interaction modes should give way to more modern, and fun ways. With the advancement of virtual reality (VR), robotics, AI and natural language interaction, there is potential to make human-robot interaction (HRI) much more fun and interesting, while also discovering new and important ways to learn about robotics and AI. The project envisages a gamification of this human-robot interaction, where two human users play a version of "Monkey in the Middle" (or, "Piggy in the Middle") game completely in VR, throwing a ball to each other using the VR controllers. The <i>monkey / piggy</i> in the middle is a virtual robot manipulator (e.g., Franka Emika Panda), which uses a pad in its hand to try and intercept the ball! If the robot indeed manages to intercept, the game can either reset or change to the next level, where one of the users now comes to the middle, i.e., becomes the <i>monkey / piggy</i>, and the other user is now playing with the robot itself. This project will allow the group to explore and learn about Unity/Unreal VR development, multiplayer gaming & networking, robot simulation, motion planning, object tracking, AI behaviour design, human-robot interaction (HRI),	

¹ An overview of different research areas in the school of computer science can be found [here](#).

etc. The project allows for game play design, and for levels of complexity in implementation, e.g., (i) the robot may simply move towards the tracked ball in a reactive way, or (ii) be more predictive in its actions by estimating player gaze, hand motion, throw direction, etc., or (iii) learn the passing patterns and player strategies during game play using AI / ML. In other scenarios, there can be a third user who is controlling the robot directly instead of the robot moving autonomously.

As you may see, there are several different components that need to work together here, and one of the challenges is the integration of these components, such as, robot motion control, virtual reality environment, game play, AI, etc. The project will help advance the state-of-the-art in many aspects of HRI that remain difficult in research, e.g., making HRI more natural, adaptability of robot actions to user preferences, and demonstrating that HRI can be fun!

Requirements

The group must provide:

A functional HRI interface in VR for the multi-person game play and the virtual robot manipulator, executing the virtual "monkey in the middle" game using VR controllers. The interface should allow at least 2 users to play the game.

Features:

- Virtual reality interaction
- Different robot motion profiles
 - o Different ways for the robot to intercept the ball
- Multi (at least 2) player game-play implementation

The system requires front-end (interface and VR interaction) as well as backend (robot motion programming, AI) development. The system is meant to be a gamified exploration of enhancing HRI experiences for human users. The group may explore beyond these features on how to make the interface more attractive for the users on showcase day by making this into a competition with a scoreboard, session winners, etc., to promote user interest.

Skills

Highly desirable:

VR interface & game development, Robot programming, frontend and backend development, pair programming, AI / ML

Desirable:

Project management, developing for users, interest in VR and robotics

Submission / Award information

Date of Submission:

September 2026

Date of Pitch:

October 2026

Notification of award:

October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.